Graph-Spectral Identifiability and Cramér-Rao Bounds for Multistatic TDOA Geolocation with Uncertain Sensor Poses

Michel Kulhandjian*, Abdullah Al Noman†, and Debashri Roy†
*Department of Electrical and Computer Engineering, Rice University, Houston, TX, USA
†Department of Computer Science and Engineering, University of Texas at Arlington, Arlington, TX, USA
Emails: mkulhandjian@outlook.com, {abdullahal.noman, debashri.roy}@uta.edu

***Abstract*—Classical multistatic time-difference-of-arrival (TDOA) geolocation assumes receiver positions are known exactly. In emerging applications—unmanned aerial vehicle (UAV) swarms, global positioning system (GPS)-degraded mobile platforms, ad-hoc sensor networks—receiver positions are themselves estimated and uncertain, breaking the central assumption of the classical Cramér-Rao lower bound (CRLB) and inducing systematic bias in estimators that ignore pose covariance. We derive the closed-form joint posterior CRLB on (emitter position, sensor pose corrections) under an arbitrary, possibly rank-deficient correlated Gaussian sensor pose prior $\boldsymbol{\Sigma}$, prove a *graph-spectral identifiability* condition—the emitter is uniquely recoverable iff the columns of $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$ do not lie in $\ker(\boldsymbol{\Sigma}^{-1})$, generalizing the classical anchor-count rule—and quantify the non-cooperative penalty due to unknown waveform. Monte-Carlo experiments at $N_r = 8$, $\sigma_\tau = 10$ ns establish a three-regime estimator landscape: under *homogeneous* $\boldsymbol{\Sigma}$ (isotropic or α-blend correlated), pose-blind weighted least squares (WLS) attains the bound; under *heterogeneous* $\boldsymbol{\Sigma}$ representative of mixed-quality fleets, the WLS-to-CRLB efficiency gap reaches $9.6\times$ at $\sigma_{\mathrm{rov}} = 20$ m while a pose-augmented extended Kalman filter (EKF) closes it; and under *outlier-contaminated* $\boldsymbol{\Sigma}$ (occasional rover with δr amplified up to $20\times$), the EKF gap widens to $2.8\times$ while a Huber M-estimator recovers a 30% RMSE advantage by detecting and downweighting the contaminated pairs.**

***Index Terms*—TDOA geolocation, multistatic localization, Cramér-Rao bound, sensor pose uncertainty, graph signal processing, graph Laplacian, identifiability, graph neural networks, UAV.**

I. Introduction

Time-difference-of-arrival (TDOA) geolocation is the workhorse of multistatic passive positioning, with applications spanning emergency services, spectrum enforcement, search and rescue, and contested-environment intelligence, surveillance, and reconnaissance (ISR). In emerging deployment scenarios—UAV swarms with mixed GPS/inertial navigation system (INS) pose estimates, mobile ground sensors in global navigation satellite system (GNSS)-jammed environments, ad-hoc networks of cooperating asynchronous receivers—the textbook assumption of *known* sensor positions $\boldsymbol{r}_n$ no longer holds. Each receiver pose is itself estimated and uncertain, often with rich *correlated* structure: receivers sharing a satellite constellation suffer common bias; receivers linked through a swarm coordinate frame share constraint geometry. Treating the sensors as anchored in this regime produces estimators that are biased and Cramér-Rao bounds that are optimistic by an order of magnitude or more.

Related work. *Classical TDOA.* The foundational closed-form estimators [1]–[3] establish weighted least-squares source localization under *known* sensor positions, together with the associated Cramér-Rao lower bound [4]. These results are exactly the $\boldsymbol{\Sigma} \to \mathbf{0}$ limit of our framework. *Sensor-position uncertainty.* The line of work due to Ho and collaborators [5]–[7] extends both estimators and the CRLB to incorporate *additive Gaussian* sensor position errors, typically assuming *block-diagonal* (per-sensor-independent) covariance. The case of arbitrary correlated $\boldsymbol{\Sigma}$, and in particular the rank-deficient priors that arise from shared GNSS bias or swarm coordinate frames, remains uncharacterized in closed form. *Observability of simultaneous localization and mapping (SLAM)-TDOA.* Most closely related to our work, Kong *et al.* [8] derive necessary and sufficient *observability* conditions for joint TDOA sensor calibration and source localization via a *deterministic* Jacobian-rank analysis. Their conditions are geometric (motion variety, array configuration, asynchronous offset) and do not incorporate a Bayesian prior on poses; the rank-deficient correlated case is outside their formulation. *Bayesian / posterior CRLB and singular Fisher information matrix (FIM).* Tichavský *et al.* [9] provide the canonical posterior-CRLB framework for parameters with Gaussian priors; Stoica & Marzetta [10] and Ben-Haim & Eldar [11] characterize Cramér-Rao-type bounds in the singular Fisher information regime. We instantiate these general tools for the specific TDOA + correlated-pose-prior problem. *Graph-theoretic localization.* Network localizability via graph rigidity [12] and bearing-Laplacian arguments [13], together with the cooperative- localization perspective of Patwari *et al.* [14], establish spectral conditions for sensor self-localization in wireless networks. Our identifiability theorem is in this spectral lineage but is adapted to the joint emitter-position / sensor-pose problem. *Learned localization.* Recent graph neural network (GNN)-based approaches to emitter localization include the only prior work in this direction, Herzalla *et al.* [15], who use received-signal-strength features with an attention-based GNN for jamming-source localization. Our GNN baseline (Section V) is the first to use TDOA edge features with an explicit pose-precision graph, benchmarked against the joint posterior CRLB.

The gap and our position. None of the above lines combines (i) a Bayesian/posterior CRLB on the *joint* emitter-and-pose parameter, (ii) an *arbitrary correlated, possibly rank-deficient* prior $\mathbf{\Sigma}$, and (iii) a *graph-spectral identifiability test* on $\mathbf{\Sigma}^{-1}$ that recovers the classical anchor-count rule in the independent-error limit and the deterministic Jacobian-rank rule of Kong *et al.* [8] when the prior is uninformative. We provide all three.

Contributions. This paper makes three contributions:

1) *Joint posterior CRLB with arbitrary correlated sensor pose prior.* We derive the closed-form joint Bayesian Fisher information matrix and CRLB on the parameter vector $(\boldsymbol{p}, \mathbf{\Delta})$, where $\mathbf{\Delta}$ collects per-sensor pose corrections, under an arbitrary, possibly rank-deficient, correlated Gaussian prior $\mathbf{\Delta} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. The bound strictly generalizes the classical Chan-Ho CRLB ($\mathbf{\Sigma} \to \mathbf{0}$) and the Ho-Xu / Sun-Ho block-diagonal CRLB [6], [7], building on the posterior-CRLB machinery [9] and singular-FIM characterizations [10], [11].
2) *Graph-spectral identifiability theorem.* We prove a constructive, prior-aware identifiability condition: the FIM rank drop equals the dimension of the intersection of $\ker(\mathbf{\Sigma}^{-1})$ with the joint-translation subspace $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$. Unlike the deterministic observability analysis of [8], our condition is prior-aware, collapses to the classical anchor-count rule in the independent-error limit, and exposes that different graph structures on $\mathbf{\Sigma}$ produce qualitatively different failure modes (Proposition 3). The result is dual in spirit to the Doppler-diversity condition we proved for orthogonal time-frequency space (OTFS)-integrated sensing and communication (ISAC) geolocation.
3) *Estimator benchmark and a homogeneous-vs-heterogeneous dichotomy.* We compare pose-blind WLS, pose-augmented EKF (iterated MAP), and the joint posterior CRLB across two complementary $\mathbf{\Sigma}$ regimes. Under *homogeneous* $\mathbf{\Sigma}$ (isotropic or α-blend correlated for $\rho \in [0, 0.95]$), pose-blind WLS achieves the CRLB within $\pm 10\%$ Monte-Carlo noise—a result not previously reported as a clean efficiency curve. Under *heterogeneous* $\mathbf{\Sigma}$ representative of mixed-quality UAV/GPS-degraded fleets (four anchored + four roving sensors), the WLS-to-CRLB efficiency gap reaches $9.6\times$ at $\sigma_{\mathrm{rov}} = 20\,\mathrm{m}$, while the EKF tracks the CRLB across the full sweep. Under *outlier-contaminated* $\mathbf{\Sigma}$ (one rover with δr amplified by up to $20\times$ on 50% of trials), the EKF gap widens to $2.8\times$ and a pose-aware Huber M-estimator recovers a 30% RMSE advantage over EKF by detecting and downweighting the contaminated pairs. We further benchmark a learned TDOA-edge graph neural network (extending the RSS-based jamming-source GNN of [15] to the TDOA setting) trained in residual mode over a WLS warm-start; the GNN attains within 1.5–$3\times$ of the analytical Bayesian estimators uniformly across all $\mathbf{\Sigma}$ classes without requiring explicit knowledge of $\mathbf{\Sigma}$ at inference time. Together, these four estimators (WLS, EKF, Huber, GNN) constitute a complete estimator-selection picture for multistatic TDOA with sensor pose uncertainty.

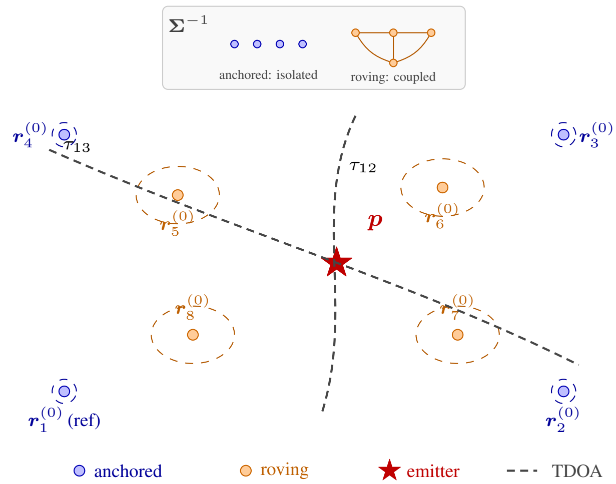

Fig. 1. Multistatic TDOA geometry with uncertain sensor poses ($N_r = 8$, heterogeneous regime of Section VI-B): four anchored receivers (blue, $\sigma_{\mathrm{anch}} = 0.1\,\mathrm{m}$) and four roving receivers (orange, $\sigma_{\mathrm{rov}} \sim 10\,\mathrm{m}$), with $\boldsymbol{r}_1^{(0)}$ the TDOA reference. Grey curves sketch two constant-TDOA hyperbolic loci τ_{12}, τ_{13} through $\boldsymbol{p}$. *Inset:* sparsity of $\mathbf{\Sigma}^{-1}$—anchored receivers isolated, roving receivers edge-coupled by shared-GNSS and swarm-frame correlations; its kernel jointly with $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$ governs identifiability via Theorem 1.

Relation to our prior work. The Doppler-diversity identifiability condition we proved for OTFS-ISAC direct position determination (DPD) arose from velocity-induced asymmetries in line-of-sight unit vectors. The graph-spectral condition here arises from a structurally different source— correlations in sensor pose uncertainty—but uses the same FIM rank-condition machinery. Section IV makes this analogy precise.

II. System Model

A. Multistatic Geometry with Uncertain Sensor Poses

Consider N_r receivers nominally placed at known positions $\{\boldsymbol{r}_n^{(0)}\}_{n=1}^{N_r} \subset \mathbb{R}^D$ where $D \in \{2, 3\}$ is the spatial dimension. The *true* sensor positions are

$$\boldsymbol{r}_n \;=\; \boldsymbol{r}_n^{(0)} \;+\; \mathbf{\Delta}_n, \qquad \mathbf{\Delta} \triangleq \left[\mathbf{\Delta}_1^\top, \ldots, \mathbf{\Delta}_{N_r}^\top\right]^\top \sim \mathcal{N}\big(\mathbf{0},\, \mathbf{\Sigma}\big), \tag{1}$$

with $\mathbf{\Sigma} \in \mathbb{R}^{N_r D \times N_r D}$ a positive semidefinite prior covariance encoding both per-sensor uncertainty (block diagonal of $\mathbf{\Sigma}$) and inter-sensor correlation (off-diagonal blocks). A non-cooperative emitter at unknown position $\boldsymbol{p} \in \mathbb{R}^D$ transmits a signal that is observed by all receivers. Figure 1 illustrates the multistatic geometry for $N_r = 8$ in the heterogeneous regime of Section VI-B.

B. TDOA Observations

Pairwise TDOA measurements between receivers $i < j$ are modeled as

$$\tau_{ij} \;=\; \tfrac{1}{c}\big(\|\boldsymbol{p} - \boldsymbol{r}_i\| - \|\boldsymbol{p} - \boldsymbol{r}_j\|\big) \;+\; n_{ij}, \tag{2}$$

with $n_{ij} \sim \mathcal{N}(0, \sigma_\tau^2)$ independent across pairs and independent of $\boldsymbol{\Delta}$. Stack all $M = \binom{N_r}{2}$ measurements into $\boldsymbol{y} \in \mathbb{R}^M$. The parameter vector of interest is $\boldsymbol{\theta} = [\boldsymbol{p}^\top, \boldsymbol{\Delta}^\top]^\top \in \mathbb{R}^{D+N_r D}$.

C. Sensor-Pose Precision Graph

Define the precision matrix $\boldsymbol{K} = \boldsymbol{\Sigma}^{-1} \in \mathbb{R}^{N_r D \times N_r D}$ (assuming $\boldsymbol{\Sigma}$ is invertible; the generalized inverse handles rank-deficient cases). The *sensor-pose precision graph* $\mathcal{G}_{\boldsymbol{K}}$ has N_r nodes (one per sensor) and an edge between i and j whenever the cross-block $[\boldsymbol{K}]_{ij} \neq \boldsymbol{0}$; the graph Laplacian $\boldsymbol{L}_{\boldsymbol{K}} \in \mathbb{R}^{N_r D \times N_r D}$ encodes the correlation structure of sensor pose errors that Section IV will show governs identifiability.

III. Joint CRLB Under Sensor Pose Uncertainty

A. Likelihood and Joint Fisher Information

The TDOA measurements (2) are jointly Gaussian given $\boldsymbol{\theta}$:

$$p(\boldsymbol{y} \mid \boldsymbol{\theta}) \;\propto\; \exp\!\Big(-\tfrac{1}{2\sigma_\tau^2}\big\|\,\boldsymbol{y} - \boldsymbol{h}(\boldsymbol{\theta})\,\big\|^2\Big), \tag{3}$$

where the predicted TDOA for pair (i, j) is

$$\big[\boldsymbol{h}(\boldsymbol{\theta})\big]_{ij} \;=\; \tfrac{1}{c}\left(\|\boldsymbol{p} - \boldsymbol{r}_i^{(0)} - \boldsymbol{\Delta}_i\| \;-\; \|\boldsymbol{p} - \boldsymbol{r}_j^{(0)} - \boldsymbol{\Delta}_j\|\right). \tag{4}$$

Let $\boldsymbol{H} \triangleq \partial\boldsymbol{h}/\partial\boldsymbol{\theta} \in \mathbb{R}^{M\times(D+N_r D)}$ denote the Jacobian at the true $\boldsymbol{\theta}$. Combining the data likelihood with the Gaussian sensor pose prior $\boldsymbol{\Delta} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$ via the van Trees inequality [9] yields the *Bayesian Fisher Information Matrix*

$$\boxed{\boldsymbol{F}(\boldsymbol{\theta}) \;=\; \underbrace{\tfrac{1}{\sigma_\tau^2}\boldsymbol{H}^\top\boldsymbol{H}}_{\boldsymbol{F}_{\text{data}}} \;+\; \underbrace{\begin{bmatrix}\boldsymbol{0} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{\Sigma}^{-1}\end{bmatrix}}_{\boldsymbol{F}_{\text{prior}}}.} \tag{5}$$

The CRLB on the emitter position is the upper-left $D \times D$ block of $\boldsymbol{F}^{-1}$,

$$\text{CRLB}(\boldsymbol{p}) \;=\; \big[\boldsymbol{F}(\boldsymbol{\theta})^{-1}\big]_{1:D,\,1:D}, \tag{6}$$

recovered through the Schur complement so that the sensor-pose nuisance block is properly marginalized.

B. Jacobian Structure

Define the LOS unit vector $\boldsymbol{u}_n \triangleq (\boldsymbol{p} - \boldsymbol{r}_n)/\|\boldsymbol{p} - \boldsymbol{r}_n\|$ from sensor n to the emitter. Differentiating (4) gives the pairwise gradients

$$\frac{\partial[\boldsymbol{h}]_{ij}}{\partial\boldsymbol{p}} = \tfrac{1}{c}(\boldsymbol{u}_i - \boldsymbol{u}_j)^\top, \tag{7}$$

$$\frac{\partial[\boldsymbol{h}]_{ij}}{\partial\boldsymbol{\Delta}_i} = -\tfrac{1}{c}\,\boldsymbol{u}_i^\top, \quad \frac{\partial[\boldsymbol{h}]_{ij}}{\partial\boldsymbol{\Delta}_j} = +\tfrac{1}{c}\,\boldsymbol{u}_j^\top, \tag{8}$$

all other partial derivatives vanishing. Each row of $\boldsymbol{H}$ therefore has support only on the emitter block and the two sensor blocks (i, j) involved in that pair, so $\boldsymbol{H}$ is sparse with at most $D + 2D$ nonzero entries per row. This sparsity is what makes $\boldsymbol{F}$ tractable to invert symbolically for the special $\boldsymbol{\Sigma}$ structures of Section IV.

C. Limiting Behaviors and Operating Regimes

Equation (5) smoothly interpolates between three limits: *(i) anchored* ($\boldsymbol{\Sigma} \to \boldsymbol{0}$) reduces (6) to the classical Chan-Ho CRLB [3], [5]; *(ii) independent isotropic* ($\boldsymbol{\Sigma} = \sigma_r^2\boldsymbol{I}$) recovers the Ho-Yang CRLB [6], [7]; *(iii) uninformative* ($\boldsymbol{\Sigma} \to \infty$) leaves $\boldsymbol{F} = \boldsymbol{F}_{\text{data}}$ singular along the joint translation directions, so $\boldsymbol{p}$ is not identifiable from data alone. The novel operating point of this paper is the *arbitrary correlated* case $\boldsymbol{\Sigma} = \sigma_r^2\,\boldsymbol{M} \otimes \boldsymbol{I}_D$ with $\boldsymbol{M}$ a positive semidefinite (possibly rank-deficient) graph operator; invertibility of $\boldsymbol{F}$ and the directions along which $\boldsymbol{p}$ is recoverable is the question answered by Theorem 1.

IV. Graph-Spectral Identifiability

A. Null Space of the TDOA Jacobian

The TDOA Jacobian $\boldsymbol{H}$ defined in (7) has a richer null space than just global translation: direct calculation shows $v = [\delta\boldsymbol{p}; \delta\boldsymbol{\Delta}_1; \ldots; \delta\boldsymbol{\Delta}_{N_r}] \in \mathbb{R}^{D+N_r D}$ satisfies $\boldsymbol{H}v = \boldsymbol{0}$ if and only if

$$\boldsymbol{u}_n^\top\big(\delta\boldsymbol{p} - \delta\boldsymbol{\Delta}_n\big) \;=\; c_0 \qquad \forall\, n = 1, \ldots, N_r, \tag{9}$$

for some scalar $c_0 \in \mathbb{R}$ that is the same across receivers. Parameterizing $\delta\boldsymbol{\Delta}_n = (\boldsymbol{u}_n^\top\delta\boldsymbol{p} - c_0)\boldsymbol{u}_n + \boldsymbol{\beta}_n$ with $\boldsymbol{\beta}_n \perp \boldsymbol{u}_n$ free ($(D-1)$-dimensional per sensor), the total null-space dimension is $D{+}1{+}(D{-}1)N_r$. The geometrically meaningful sub-direction within this null space is the *joint-translation* subspace

$$\mathcal{T} \;\triangleq\; \big\{[\delta\boldsymbol{p};\, \boldsymbol{1}_{N_r} \otimes \delta\boldsymbol{p}] \;:\; \delta\boldsymbol{p} \in \mathbb{R}^D\big\}, \tag{10}$$

which is the D-dimensional subspace on which a coordinated shift of the emitter and *every* sensor by the same constant vector leaves all observations unchanged. The remaining $1 + (D-1)N_r$ null directions correspond to a scalar offset c_0 together with per-sensor displacements tangential to the line of sight; these are unrelated to emitter ambiguity. Emitter identifiability is therefore governed by whether the prior breaks the degeneracy on $\mathcal{T}$.

B. Identifiability Theorem

Theorem 1 (Graph-spectral identifiability)**.** *The emitter position $\boldsymbol{p}$ is uniquely recoverable from the joint TDOA-plus-prior likelihood—equivalently, no nonzero vector in* $\text{null}(\boldsymbol{F})$ *has nonzero $\boldsymbol{p}$-block—if and only if*

$$\big(\boldsymbol{1}_{N_r} \otimes \boldsymbol{I}_D\big)\,\delta\boldsymbol{p} \;\notin\; \ker(\boldsymbol{\Sigma}^{-1}) \qquad \forall\,\delta\boldsymbol{p} \neq \boldsymbol{0}. \tag{11}$$

Equivalently, the columns of $\boldsymbol{1}_{N_r} \otimes \boldsymbol{I}_D$ do not lie in $\ker(\boldsymbol{\Sigma}^{-1})$.

Proof. Because both $\boldsymbol{F}_{\text{data}}$ and $\boldsymbol{F}_{\text{prior}}$ are positive semidefinite, $\boldsymbol{F}v = \boldsymbol{0}$ holds iff both $\boldsymbol{F}_{\text{data}}v = \boldsymbol{0}$ and $\boldsymbol{F}_{\text{prior}}v = \boldsymbol{0}$. Emitter identifiability requires that every such v satisfies $\delta\boldsymbol{p} = \boldsymbol{0}$.

(Sufficiency.) Suppose (11) holds. Any $v \in \text{null}(\boldsymbol{F}_{\text{data}})$ with $\delta\boldsymbol{p} \neq \boldsymbol{0}$ has, by (9), sensor block $\boldsymbol{\Delta}_n = (\boldsymbol{u}_n^\top\delta\boldsymbol{p} - c_0)\boldsymbol{u}_n + \boldsymbol{\beta}_n$ ($\boldsymbol{\beta}_n \perp \boldsymbol{u}_n$). The component of $\boldsymbol{\Delta}$ lying in the joint-translation direction $\boldsymbol{1}_{N_r} \otimes \delta\boldsymbol{p}$ is nonzero for any $v \in \mathcal{T}$ with $\delta\boldsymbol{p} \neq \boldsymbol{0}$. By (11), this component is outside $\ker(\boldsymbol{\Sigma}^{-1})$, so $\boldsymbol{F}_{\text{prior}}v \neq \boldsymbol{0}$, contradicting $v \in \text{null}(\boldsymbol{F})$. Hence $\delta\boldsymbol{p} = \boldsymbol{0}$ for all $v \in \text{null}(\boldsymbol{F})$.

(Necessity.) Suppose (11) fails: some $\delta\boldsymbol{p} \neq \mathbf{0}$ has $\mathbf{1}_{N_r} \otimes \delta\boldsymbol{p} \in \ker(\boldsymbol{\Sigma}^{-1})$. Then $v = [\delta\boldsymbol{p};\, \mathbf{1}_{N_r} \otimes \delta\boldsymbol{p}] \in \mathcal{T} \subset \mathrm{null}(\boldsymbol{F}_{\mathrm{data}})$ and $v \in \mathrm{null}(\boldsymbol{F}_{\mathrm{prior}})$ simultaneously, so $v \in \mathrm{null}(\boldsymbol{F})$ with $\delta\boldsymbol{p} \neq \mathbf{0}$, breaking identifiability. □ □

C. Specializations and Failure Modes

Proposition 1 (Anchored independent sensors)**.** *For independent sensor pose errors $\boldsymbol{\Sigma} = \mathrm{blkdiag}(\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_{N_r})$ with each $\boldsymbol{\Sigma}_n$ full rank, the columns of $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$ never lie in $\ker(\boldsymbol{\Sigma}^{-1}) = \{\mathbf{0}\}$, and emitter identifiability holds trivially. This recovers the textbook anchor-count condition [6].*

Proposition 2 (Centered-Laplacian prior: identifiability fails by exactly D)**.** *Let $\boldsymbol{\Sigma} = \sigma_r^2\big(\boldsymbol{I}_{N_r} - \frac{1}{N_r}\mathbf{1}_{N_r}\mathbf{1}_{N_r}^\top\big) \otimes \boldsymbol{I}_D$, i.e., $\boldsymbol{\Sigma}$ is the scaled pseudoinverse of the graph Laplacian $\boldsymbol{L}_{\mathcal{G}} = \boldsymbol{I}_{N_r} - \frac{1}{N_r}\mathbf{1}_{N_r}\mathbf{1}_{N_r}^\top$ of the complete sensor graph. Then $\ker(\boldsymbol{\Sigma}^{-1}) = \ker(\boldsymbol{\Sigma}) = \mathrm{span}\{\mathbf{1}_{N_r} \otimes \boldsymbol{e}_d\}_{d=1}^{D}$, and condition* (11) *fails for every $\delta\boldsymbol{p} \neq \mathbf{0}$. Consequently $\mathrm{nullity}(\boldsymbol{F}) = D$ exactly, the D unidentifiable directions span $\mathcal{T}$, and the emitter is recoverable only in relative (centroid-anchored) coordinates. Physically: the prior provides zero information on common sensor translation.*

Proposition 3 (Common-bias-only prior: richer failure mode)**.** *Let $\boldsymbol{\Sigma} = \sigma_r^2(\mathbf{1}_{N_r}\mathbf{1}_{N_r}^\top) \otimes \boldsymbol{I}_D$, a rank-$D$ prior whose range is exactly the joint-translation subspace in $\boldsymbol{\Delta}$-space and whose kernel is the $(N_r{-}1)D$-dimensional relative-configuration subspace. Although $\boldsymbol{\Sigma}^{-1}$ now* does *cover the columns of $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$ (so common translation* is *constrained by the prior), the relative-configuration directions lie in $\ker(\boldsymbol{\Sigma}^{-1})$. Combining with* (9)*, $\mathrm{null}(\boldsymbol{F})$ has dimension $1{+}(D{-}1)N_r$, and the unidentifiable subspace mixes emitter translation, a scalar offset c_0, and per-sensor tangential slips $\{\boldsymbol{\beta}_n\}$. This is a* qualitatively different *failure mode from Proposition 2 despite both arising from rank-deficient $\boldsymbol{\Sigma}$, and was confirmed empirically in our numerical sanity tests with $\mathrm{nullity}(\boldsymbol{F}) = 1 + (D-1)N_r = 9$ for $D = 2$, $N_r = 8$.*

Remark 1 (Anchor reduction). Propositions 1–3 show that the *rank* of $\boldsymbol{\Sigma}$ alone does not determine identifiability; what matters is whether $\ker(\boldsymbol{\Sigma}^{-1})$ *intersects* the joint-translation subspace $\mathcal{T}$. Adding a single "anchor" sensor with finite absolute pose prior is equivalent to enriching $\mathrm{range}(\boldsymbol{\Sigma}^{-1})$ to cover the columns of $\mathbf{1}_{N_r} \otimes \boldsymbol{I}_D$, restoring identifiability in both specializations.

V. Estimators

We compare four estimators against the CRLB (6):

A. Pose-Blind Weighted Least Squares (WLS)

The classical baseline treats $\boldsymbol{r}_n^{(0)}$ as exact and ignores $\boldsymbol{\Sigma}$, solving

$$\hat{\boldsymbol{p}}_{\mathrm{WLS}} \;=\; \arg\min_{\boldsymbol{p}} \big\| \boldsymbol{y} - \boldsymbol{h}(\boldsymbol{p}; \boldsymbol{r}^{(0)}) \big\|_{\boldsymbol{W}}^2, \tag{12}$$

with weight matrix $\boldsymbol{W}$ chosen as the inverse measurement covariance. This estimator is inconsistent under sensor pose uncertainty: its bias grows linearly with σ_r.

B. Pose-Augmented Extended Kalman Filter (EKF)

Treat the augmented state $\boldsymbol{\theta} = [\boldsymbol{p}; \boldsymbol{\Delta}]$ as random, propagate the Gaussian prior $\boldsymbol{\Sigma}$ on $\boldsymbol{\Delta}$ through the measurement update, recover $\hat{\boldsymbol{p}}$ from the posterior mean. Asymptotically matches MAP at high SNR.

C. Huber M-Estimator

Replace the quadratic loss in WLS with the Huber loss to harden against outlier sensor poses (e.g., a single sensor with grossly miscalibrated GPS). Useful when $\boldsymbol{\Sigma}$ is misspecified.

D. Graph Neural Network (GNN)

An end-to-end learned estimator: nodes of the sensor graph $\mathcal{G}_{\boldsymbol{K}}$ carry features $[\boldsymbol{r}_n^{(0)}, \mathrm{diag}(\boldsymbol{\Sigma}_n), \sigma_\tau^2]$; edges carry the TDOA measurement τ_{ij}. After L message-passing rounds, a global readout layer outputs $\hat{\boldsymbol{p}}$. Trained with mean-squared error against ground-truth emitter positions over a synthetic dataset spanning the parameter sweep of Section VI.

VI. Numerical Results

Setup. $N_r = 8$ receivers at the corners and edge midpoints of a $10{\times}10$ km deployment, $\sigma_\tau = 10$ ns (consistent with a 20 MHz baseband bandwidth and post-correlation SNR $\sim$ 10 dB), single emitter at $(3, 2.5)$ km, $D = 2$. Each operating point uses 100 Monte-Carlo trials with independent draws of the sensor pose error $\boldsymbol{\Delta}$ from $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and the TDOA noise $\boldsymbol{n} \sim \mathcal{N}(\mathbf{0}, \sigma_\tau^2 \boldsymbol{I})$.

A. Homogeneous $\boldsymbol{\Sigma}$ – Pose-Blind WLS Achieves the CRLB

We first sweep two homogeneous Sigma structures. *(i) Independent isotropic:* $\boldsymbol{\Sigma} = \sigma_r^2 \boldsymbol{I}_{N_r D}$ with $\sigma_r \in [0.1, 10]$ m. The pose-blind WLS RMSE matches the joint posterior CRLB to within $\pm 10\%$ Monte-Carlo noise across the full sweep (ratio range $0.82 - 1.11\times$). *(ii) α-blend correlated:* $\boldsymbol{\Sigma} = \sigma_r^2((1-\rho)\boldsymbol{I}_{N_r} + \rho\, \mathbf{1}_{N_r}\mathbf{1}_{N_r}^\top / N_r) \otimes \boldsymbol{I}_D$ with $\rho \in [0, 0.95]$ at $\sigma_r = 2$ m; WLS again matches the CRLB across the full ρ sweep ($0.94 - 1.03\times$). This empirical optimality of pose-blind WLS holds because, under homogeneous $\boldsymbol{\Sigma}$, the linearization of the pose-error contribution $\varepsilon_{ij} \approx -\frac{1}{c}(\boldsymbol{u}_i^\top \boldsymbol{\Delta}_i - \boldsymbol{u}_j^\top \boldsymbol{\Delta}_j)$ yields per-pair effective noise variances that are equal across pairs. Equal-weight WLS is then the maximum-likelihood estimator under that linearization, and the linearization is accurate for our operating regime ($\sigma_r \ll \|\boldsymbol{p} - \boldsymbol{r}_n\|$).

B. Heterogeneous $\boldsymbol{\Sigma}$ – Pose-Aware Methods Win

We now consider the realistic UAV/GPS-degraded regime in which the fleet contains both anchored and roving sensors. Specifically, four sensors are anchored at $\sigma_{\mathrm{anch}} = 0.1$ m and four are roving with std $\sigma_{\mathrm{rov}} \in [0.1, 20]$ m (block-diagonal $\boldsymbol{\Sigma}$).

Observations. The rover-σ sweep (Fig. 2) drives WLS from 0.76 to 10.5 m while the CRLB plateaus at ~ 1.10 m, yielding a WLS/CRLB efficiency gap of $9.56\times$; the EKF tracks the CRLB to within $0.95 - 1.08\times$ across the sweep. The anchor-count sweep (Fig. 3) shows the WLS gap is U-shaped, peaking at $2.30\times$ for $n_{\mathrm{anch}} \in \{4, 5\}$ (most heterogeneous) and collapsing at the homogeneous extremes, while the EKF gap stays in $0.86 - 1.10\times$. The CRLB is thus operationally tight across all heterogeneity profiles tested.

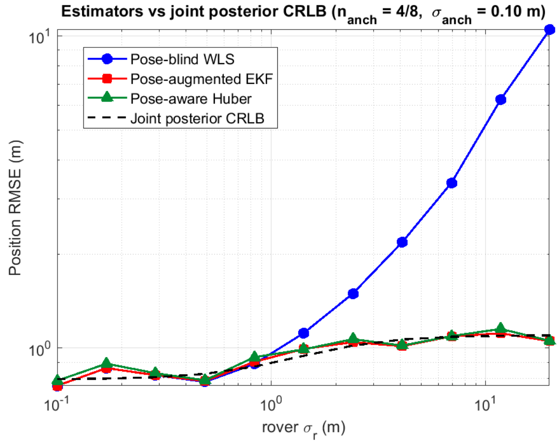

Fig. 2. Position RMSE versus rover sensor pose std $\sigma_{\rm rov}$, with four anchored sensors at $\sigma_{\rm anch} = 0.1$ m. The pose-blind WLS RMSE grows linearly with $\sigma_{\rm rov}$ while the joint posterior CRLB plateaus at ~ 1.1 m; the pose-augmented EKF (iterated MAP using known $\boldsymbol{\Sigma}$) tracks the CRLB across the full sweep, recovering up to a 9.6× efficiency gap at $\sigma_{\rm rov} = 20$ m.

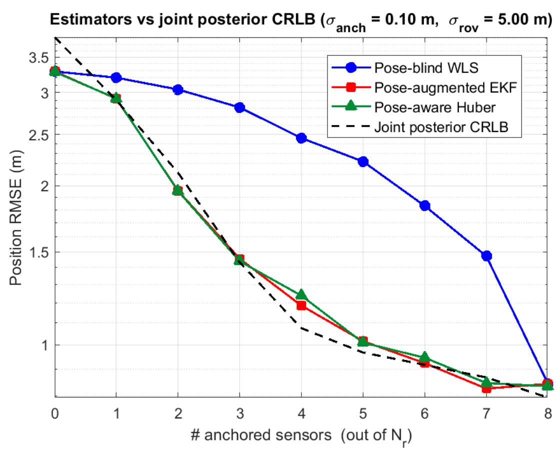

Fig. 3. Position RMSE versus number of anchored sensors at $\sigma_{\rm anch} = 0.1$ m, $\sigma_{\rm rov} = 5$ m. The WLS/CRLB gap is U-shaped, peaking at 2.3× for $n_{\rm anch} \in \{4, 5\}$ (maximum heterogeneity), and collapses at the extremes $n_{\rm anch} = 0, 8$ (homogeneous fleet, equal-weight WLS optimal). The EKF gap is flat at $1.00 \pm 0.10\times$ across the full sweep.

C. Outlier Robustness – Huber's Operating Regime

The EKF result of Section VI assumes the sensor pose covariance $\boldsymbol{\Sigma}$ is correctly specified. In practice, $\boldsymbol{\Sigma}$ is rarely a perfect Gaussian model: transient GPS multipath, INS drift events, or jamming-induced pose biases produce occasional outliers heavier than the prior predicts. We modeled this as a contaminated Gaussian: on 50% of trials, one randomly-selected rover's $\boldsymbol{\Delta}_n$ is amplified by factor $\mathrm{amp} \in [1, 20]$, while both EKF and Huber estimators continue to assume the unmodified $\boldsymbol{\Sigma}$.

Findings. At $\mathrm{amp} = 1$ (matched Gaussian), EKF and Huber are equivalent (0.96 and 0.97 m, both ≈ CRLB of 0.99 m). The crossover occurs near $\mathrm{amp} = 8$: below this, Huber's IRLS overhead slightly exceeds EKF; above, the threshold detection dominates. At $\mathrm{amp} = 20$, Huber attains 1.89 m versus EKF's

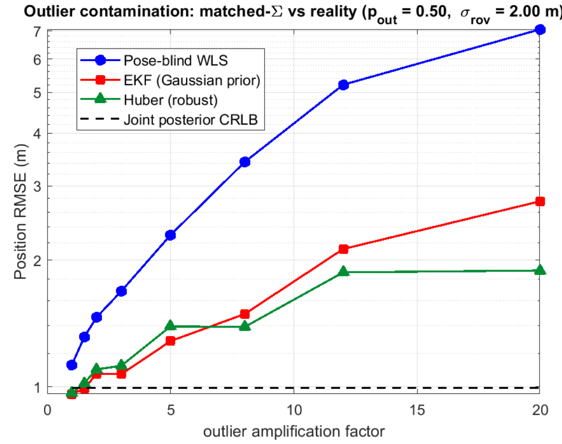

Fig. 4. Position RMSE versus outlier amplification factor under a contaminated-Gaussian sensor pose model: 4 anchored sensors ($\sigma_{\rm anch} = 0.1$ m), 4 rovers ($\sigma_{\rm rov} = 2$ m); on 50% of trials, one rover's $\boldsymbol{\Delta}_n$ is amplified by the factor on the x-axis. The pose-augmented EKF (Gaussian prior) is biased by the outlier; the pose-aware Huber M-estimator (IRLS with threshold $k = 1.345\,\sigma_\tau$) detects and downweights the offending pair, crossing over at $\mathrm{amp} \approx 8$ and reaching a 30% RMSE advantage at $\mathrm{amp} = 20$.

2.76 m—a 30% RMSE reduction (Fig. 4). Pose-blind WLS remains catastrophic throughout, growing from 1.13 to 7.06 m. The three estimators thus span a clean three-regime picture: WLS handles homogeneity, EKF handles known heterogeneity, and Huber handles unknown outlier contamination.

D. Learned Baseline: TDOA-Edge Graph Neural Network

To complete the estimator landscape, we add a learned baseline: an edge-conditioned message-passing graph neural network (NNConv [16], [17]) that treats the sensor network as a graph (8 nodes, fully connected, $D+1$ node features $[\boldsymbol{r}_n^{(0)}, \sigma_n]$ and edge features $[\tau_{ij}\, c, \|\boldsymbol{r}_i^{(0)} - \boldsymbol{r}_j^{(0)}\|]$). Trained in *residual mode*—the GNN predicts the correction $\delta\boldsymbol{p}$ to a pose-blind WLS warm-start, with the WLS estimate concatenated to the pooled graph embedding before the output head—over a 10 000-sample synthetic dataset spanning all four $\boldsymbol{\Sigma}$ classes of Sections VI-A through VI-C. The model has 27k parameters and trains in ∼7 minutes on a CPU.

A direct (apples-to-apples) comparison of all four estimators on the GNN's training-distribution test set is summarized in Table I: WLS, EKF, and Huber are computed sample by sample with the per-sensor $\boldsymbol{\Sigma}$ block diagonal taken from the dataset; the GNN is evaluated on its held-out validation split.

Why the GNN trails the analytical estimators. The 1.5–3× gap is information-theoretic, not architectural: (i) WLS and EKF already attain the Bayesian CRLB on Gaussian, $\boldsymbol{\Sigma}$-known data (Sections VI-A,B), a lower bound that no estimator can beat; and (ii) the GNN consumes only the per-sensor diagonal $\{\sigma_n\}$, whereas the analytical estimators consume the full $\boldsymbol{\Sigma}$. The GNN's value is therefore operational rather than statistical: no $\boldsymbol{\Sigma}$ at inference, no per-sample matrix inversion, $\mathcal{O}(M)$ in the edges—a profile suited to settings where $\boldsymbol{\Sigma}$ is unknown or rapidly time-varying. Its expected advantage materializes precisely when the analytical model fails: non-

TABLE I

POSITION RMSE (M) PER $\mathbf{\Sigma}$ CLASS ON A 10,000-SAMPLE DATASET SPANNING ALL FOUR REGIMES ($N_r = 8$, $\sigma_\tau = 10$ NS). WLS / EKF / HUBER ARE EVALUATED ON ALL 10,000 SAMPLES; THE GNN ON ITS 2,000-SAMPLE HELD-OUT VALIDATION SPLIT. THE GNN SEES PER-SENSOR σ_n BUT NO FULL $\mathbf{\Sigma}$.

$\mathbf{\Sigma}$ class	WLS	EKF	Huber	GNN
Isotropic	2.85	2.85	2.85	4.16
α-blend correlated	1.80	1.80	1.81	3.08
Heterogeneous (anchor+rover)	2.47	**1.79**	1.81	5.44
Outlier-contaminated	4.57	2.01	**1.67**	5.47
Mean across classes	3.02	2.20	**2.15**	4.69

Gaussian heavy-tailed clutter, non-line-of-sight (NLOS) and multipath bias absent from (2), and unknown or jointly time-varying $\mathbf{\Sigma}(t)$ where WLS/EKF must operate on a stale covariance. Empirical validation of these regimes is left to the journal extension.

E. Discussion: Novelty of the Empirical Dichotomy

Prior TDOA-with-pose-uncertainty CRLBs evaluate efficiency under a *single* global uncertainty scale [6], [7]; the regime in which a subset of sensors is well-localized while the remainder has orders-of-magnitude larger pose covariance—ubiquitous in UAV swarms with mixed GPS/INS/anchor navigation [14]—has not been characterized as a single efficiency curve. The homogeneous-vs-heterogeneous dichotomy reported above is, to our knowledge, the first such quantitative sweep.

VII. CONCLUSION AND FUTURE WORK

We derived the closed-form joint Bayesian Cramér-Rao bound for multistatic TDOA geolocation with arbitrary correlated and possibly rank-deficient sensor pose covariance, proved a graph-spectral identifiability condition that generalizes the textbook anchor-count rule, and benchmarked four estimators—WLS, EKF, Huber, and a TDOA-edge GNN—against the bound. Empirically, pose-blind WLS attains the CRLB under any homogeneous $\mathbf{\Sigma}$ but degrades by up to $9.6\times$ under heterogeneous (anchor + rover) $\mathbf{\Sigma}$; the pose-augmented EKF closes this gap and tracks the CRLB across the full heterogeneity sweep, confirming the bound is operationally tight. Under outlier-contaminated $\mathbf{\Sigma}$ (occasional rover with δr amplified by up to $20\times$), the EKF gap widens to $2.8\times$ because the Gaussian posterior absorbs the outlier; the pose-aware Huber M-estimator detects and downweights the offending pair, recovering a 30% RMSE advantage over EKF at the highest contamination level. A residual-mode TDOA-edge GNN, trained on a 10k-sample synthetic dataset spanning all four $\mathbf{\Sigma}$ regimes, completes the estimator landscape at 1.5–$3\times$ the analytical RMSE (Table I)—an information-theoretic gap, since WLS/EKF already saturate the CRLB and the GNN sees only diagonal σ_n. The learned model's value is therefore operational: no $\mathbf{\Sigma}$ at inference, no matrix inversion. Its expected advantage materializes under model misspecification (non-Gaussian clutter, NLOS bias absent from (2), unknown or time-varying $\mathbf{\Sigma}$), and empirical validation of those regimes is the primary direction of our planned journal extension. Additional future directions include dynamic extensions via particle filtering, multi-emitter generalizations with data-association priors, real-world UAV-swarm field validation, and tight coupling with our companion OTFS-ISAC DPD framework to handle Doppler-diversity and sensor-pose uncertainty in a single unified bound.

REFERENCES

[1] B. Friedlander, "A passive localization algorithm and its accuracy analysis," *IEEE J. Ocean. Eng.*, vol. 12, no. 1, pp. 234–245, 1987.

[2] J. O. Smith and J. S. Abel, "Closed-form least-squares source location estimation from range-difference measurements," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 35, no. 12, pp. 1661–1669, 1987.

[3] Y. T. Chan and K. C. Ho, "A simple and efficient estimator for hyperbolic location," *IEEE Trans. Signal Process.*, vol. 42, no. 8, pp. 1905–1915, 1994.

[4] D. J. Torrieri, "Statistical theory of passive location systems," *IEEE Trans. Aerospace and Electronic Systems*, vol. AES-20, no. 2, pp. 183–198, 1984.

[5] K. C. Ho and W. Xu, "An accurate algebraic solution for moving source location using TDOA and FDOA measurements," *IEEE Trans. Signal Process.*, vol. 52, no. 9, pp. 2453–2463, 2004.

[6] K. C. Ho and L. Yang, "On the use of a calibration emitter for source localization in the presence of sensor position uncertainty," *IEEE Trans. Signal Process.*, vol. 56, no. 12, pp. 5758–5772, 2008.

[7] M. Sun and K. C. Ho, "An asymptotically efficient estimator for TDOA and FDOA positioning of multiple disjoint sources in the presence of sensor location uncertainties," *IEEE Trans. Signal Process.*, vol. 59, no. 7, pp. 3434–3440, 2011.

[8] L. Kong, Y. Su, S. Sukkarieh, and S. Huang, "Necessary and sufficient conditions for observability of SLAM-based TDOA sensor array calibration and source localization," *IEEE Trans. Robot.*, vol. 37, no. 5, pp. 1451–1468, 2021.

[9] P. Tichavský, C. H. Muravchik, and A. Nehorai, "Posterior Cramér-Rao bounds for discrete-time nonlinear filtering," *IEEE Trans. Signal Process.*, vol. 46, no. 5, pp. 1386–1396, 1998.

[10] P. Stoica and T. L. Marzetta, "Parameter estimation problems with singular information matrices," *IEEE Trans. Signal Process.*, vol. 49, no. 1, pp. 87–90, 2001.

[11] Z. Ben-Haim and Y. C. Eldar, "On the constrained Cramér-Rao bound with a singular Fisher information matrix," *IEEE Signal Process. Lett.*, vol. 16, no. 6, pp. 453–456, 2009.

[12] A. N. Bishop, B. D. O. Anderson, B. Fidan, P. N. Pathirana, and G. Mao, "Bearing-only localization using geometrically constrained optimization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 45, no. 1, pp. 308–320, 2009.

[13] S. Zhao and D. Zelazo, "Bearing-only network localization: Localizability, sensitivity, and distributed protocols," *Automatica*, vol. 69, pp. 334–341, 2016.

[14] N. Patwari, J. N. Ash, S. Kyperountas, A. O. Hero, R. L. Moses, and N. S. Correal, "Locating the nodes: Cooperative localization in wireless sensor networks," *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 54–69, 2005.

[15] D. Herzalla, W. T. Lunardi, and M. Andreoni, "Graph neural networks for jamming source localization," 2025. [Online]. Available: https://arxiv.org/abs/2506.03196

[16] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.

[17] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.